

Dams | Quinte Conservation

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Public Notice: Herbicide Application Taking Place for Water Soldier Management control efforts for invasive Water Soldier will be taking place in the Bay of Quinte starting the week of October 5th. Residents in areas of application have been notified. Visit: <https://www.invadingspecies.com/programs/water-soldier-eradication/> to follow along with management efforts. Conservation Authority Amalgamation Update In accordance with changes to the Conservation Authorities Act, Ontario's 36 conservation authorities will amalgamate into 9 Regional Conservation Authorities effective February 1, 2027. As part of this transition, Quinte Conservation (QC), Kawartha Conservation (KC), Ganaraska Conservation Authority (GCA), Otonabee Conservation Authority (OCA), Crowe Valley Conservation (CVC), Lower Trent Conservation (LTC), and Cataraqui Conservation (CC) are preparing to merge into the Eastern Lake Ontario Regional Conservation Authority (ELORCA). Throughout this process and beyond February 1, 2027, Quinte Conservation will continue to provide and carry out all of its programs with the high level of service expected by our residents, municipal partners, and stakeholders. Our commitment to delivering exceptional programs and services across the watershed remains unchanged. If you have questions about Quinte Conservation's programs and services, or the transition to ELORCA, please contact us at info@quinteconservation.ca. Our management team will be pleased to assist you. There are many dams in the Quinte watershed. Quinte Conservation owns and operates 40 dams and there are many other privately owned dams. An interactive map of Ontario dams is available through the Ontario Ministry of Natural Resources and Forestry. Each dam owned by us provides one or more benefits. Examples include: flood management, raising low water levels, recreation, water supply, habitat creation, historic, and hydro power production. The dams were built decades ago. They need ongoing repairs to stay in good working order. We follow operating plans for each dam. The plans tell us when to operate based on time of year or water level. The plans also tell us how to operate. These plans consider all water users and the health of the lakes and rivers. Many of our dams are only operated in the spring and fall. In the fall, water levels are lowered above the dam to reduce winter ice damage and spring flooding. After the snow has melted in the spring, we operate the dams to maintain higher summer water levels on lakes and to raise low river levels during droughts. Our dams can be controlled by stop logs and panels, valves, or inflatable weirs. Some dams cannot be operated. These dams are referred to as weirs. Weirs and dams that are only operated in the spring and fall are referred to as "run of the river" dams. This means that the dams do not change the flow in the river. All of the water that flows to the dam, flows over the dam, and continues downstream. Water levels are affected by many factors, such as rainfall and snowmelt, dam operations, ice jams, beaver activity, and evaporation. Rainfall and snowmelt often increase water levels on lakes and rivers. Dam operations can raise and lower

water levels above and below the dam. Ice jams can cause a rise in water levels directly behind the ice jam. Beaver dams can cause a rise in water levels above the dam and lower water levels downstream. Our dams affect water levels immediately upstream, this is called the headpond. Water travels downstream because of gravity. If you are interested in knowing how big the headpond of a dam is, travel upstream until you see water flowing downhill (this can look like a section of rapids or water flowing over another dam). The dam does not back water up over the rapids or the dam. Some of our dams can assist in flood management by collecting and holding excess water in the dam headpond. This reduces the water flowing through the river channel during the flood. A headpond can only hold a certain amount of water before it floods upstream, or high water levels washout the dam. Extreme rainfalls and snowmelts may contain too much water for our dams to stop flooding. The headpond can only collect water that drains to it. A flood caused by local rainfall downstream cannot be protected by the dam. We strive to provide flood management with our dams, but we are limited by the size of the headpond, and the location of rainfall or snowmelt. During droughts we operate some dams to increase water levels in the river. The amount of water in the headpond determines how much water is available to assist the river. The shape of the dam outlet and the operation determines how long river levels will be improved. As a result of historic ice jam floods, dams were built to reduce the risk of ice jams. These dams do three things: The dams create a stable water level in steep areas that would otherwise have turbulent rapids. The calm water allows an ice sheet to form. The ice sheet insulates fast flowing water from frigid, cold air that can cause frazil ice to form and jam. Floating ice booms are strung across the river upstream of the dam. The ice booms hold the ice sheet and ice chunks in place to limit travel and ice jamming downstream. We own and operate one hydro-electric dam, McLeod Dam. The green energy facility started making power in 2008. The original dam was updated with two turbines and generators that harness the renewable energy which is distributed locally. The dam makes enough energy to power 400 homes. Quinte Conservation conducted a Water Control Structure Condition Assessment, which became the foundation of the Water Infrastructure Capital Asset Plan. The Plan was approved by the Quinte Conservation Executive Board and implemented in September 2021. The purpose of the assessment is to determine the priority capital projects for the next 10 years. As part of the work, each structure was inspected and recommendations were provided. The recommendations focus on how to properly maintain or upgrade the structure to ensure safe operation, public safety, and longevity. A major component of the Plan is to complete individual Dam Safety Review's (DSR) under the current Ministry of Natural Resources and Forestry guidelines (2011). The DSR is a systematic review and evaluation of all aspects of design, construction, maintenance, operation, surveillance, and other factors, processes and systems affecting a dam's safety. The studies may make recommendations that Quinte Conservation will consider and, if required, add them to the 10-year plan.

Location: Napanee River, Hamlet of Colebrook, Township of Stone Mills
Year of construction: 1836
Description: Concrete dam with stop logs and an overflow spillway
Purpose: The dam was built to provide water power for a saw and grist mill. Now the dam provides recreation, such as, swimming, boating, and limited fishing.
Operation: This dam is only operated for maintenance

Location: Outlet of Deerock Lake, flows into Partridge Creek, Addington Highlands
Year of construction: 1969
Description: Earth dam, valve and concrete spillway
Purpose: Recreation, low flow, and habitat
Operation: Water levels are lowered in the fall. Valve is closed after the snow melt. The valve is operated as needed to provide flow to the Skootamatta River in the summer.

Location: Salmon River, Village of Lonsdale, Tyendinaga Township
Year of construction: 1981 (reconstruction)
Description: Concrete weir with two stop log bays
Purpose: Built to power a mill. Now the dam provides flood and ice management, recreation, and low flow.
Operation: In early summer, logs are installed in the dam. The south dam can be operated for low flow. There are no high flow operations. In the fall, some logs are removed from the dam. Two stop logs are set above the water level for the ice sheet to form against. These logs are removed before the winter thaw.

Location: 1.2 kilometres downstream of Beaver Lake, Township of Stone Mills
Year of construction: 1975
Description: Concrete weir and two stop log bays
Purpose: Recreation, fisheries, and flow

augmentation Operation: Typically, logs are removed in the fall and replaced in the spring. Logs are removed to let water downstream when flows are low. Logs are removed if water levels get too high. Location: Outlet of Lingham Lake on Black River, Grimsthorpe Township, Municipality of Tweed Year of construction: 1925 Description: Rock crib dam, stop logs and earth saddle dams Purpose: Low flow, recreation, and fisheries Operation: Stop logs are installed after the spring freshet. During the summer months, water levels are lowered to maintain flow in the Black River. Stop logs are removed in the fall. Location: Deer Creek, O'Hara Mill Pioneer Village and Conservation Area, Madoc Township Year of construction: Reconstructed in 2010 Description: Concrete dam with stop logs and mill sluice Purpose: Recreation and historic Operation: The headpond water levels are maintained within the seasonal operating ranges. This includes removing 2 of the 5 stop logs in the fall and replacing them following the spring freshet. Location: Outlet of Second Depot Lake into Depot Creek, 11 kilometres northwest of Verona, Township of Central Frontenac Year of construction: 1958 Description: Large earth dam, two stop log bays, and a valve Purpose: Flood management, low flow, recreation, habitat, and water supply for the Town of Greater Napanee. Operation: Operated with Third Depot Lake Dam. Water levels are maintained in the operating range, with lower levels in the winter and higher levels in the summer. The dam is operated to provide flow to the Napanee River during low water. Location: Outlet of Skootamatta Lake into Skootamatta River, Township of Addington Highlands Year of construction: 1955 Description: Concrete dam, two stop log bays, an overflow spillway, and two saddle dams Purpose: Recreation, fisheries, habitat, low flow, and flood management Operation: Logs are removed in the fall. After the snow melt, logs are replaced. Logs are adjusted in the summer to maintain summer water levels and flow in the Skootamatta River. Location: Outlet of Third Depot Lake into Depot Creek, 16 kilometres northwest of Verona, Township of Central Frontenac Year of construction: 1975 Description: Large earth dam with two valves Purpose: Flood management, low flow, and recreation Operation: Operated with Second Depot Lake Dam. Water levels are maintained in the operating range, with lower levels in the winter and higher levels in the summer. The dam is operated to provide flow to the Napanee River during low water. Location: Outlet of Big Clear Lake, upstream of Arden Lake, Hamlet of Arden, Township of Central Frontenac Year of construction: Description: Concrete dam with stop logs Purpose: Water supply, and recreation Operation: Water levels are lowered in the fall and winter to prepare for the spring snow melt. Logs are placed in the dam after the snow melt. Low flow on the Salmon River is improved by making space between the logs or removing logs. High fall flows may require more logs to be removed. Location: 200 metres downstream of the outlet of Varty Lake, Township of Stone Mills Year of construction: 1972 Description: Concrete dam with stop logs Purpose: Flood management, recreation, and habitat creation Operation: Logs are removed in the fall and replaced after the snow melt. More logs may be removed during large snow melts or rainfalls. A gate valve may assist with low flow.